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SentiML ++: an extension of the SentiML sentiment annotation scheme

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ABSTRACT
The amount of opinionated data on the web has exponentially increased especially after the emergence of online social networks. To deal with these huge deluge of data, we need to have robust mechanisms that can help identify all aspects of opinion segment and support the automatic processing of opinion data. Recently, there have been a few developments made in this direction, and different sentiment annotation schemes have been proposed such as the *SentiML*, *OpinionMiningML*, and *EmotionML*. In this work, we propose *SentiML++*, an extension of SentiML with a focus on annotating opinions, and further answering aspects of the general question "who has what opinion about whom in which context?". A detailed comparison with SentiML and other existing annotation schemes is also presented. The data collection annotated with SentiML has been annotated with *SentiML++* and is available for download for further research purposes. Experiments with data collections annotated with SentiML and SentiML++ proves that SentiML++ is a significant and valuable addition to SentiML.

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1. Introduction

The World Wide Web (WWW) has experienced revolutionary changes in its nature since the time of its birth. Multi-perspective progressive developments have transformed it from a simple read-only web to a socialised intelligent web. In the start, information retrieval (IR) systems used to exploit only the relevance of the data to satisfy users' information needs; however, later on they started considering other features such as time (e.g. searching news from the last 7 days), genre (news, scholarly articles), data type (videos, images), etc., to cater for the evolving information needs of their users. An exponential increase in opinion data was observed after the emergence of online social networks. The huge and growing subsequent volumes of opinionated data on the web created an opportunity and gave birth to the concept of opinion mining.

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Opinion Mining is the process of extracting opinions from textual segments automatically (Liu, 2012). In the literature, it has commonly been referred to as Sentiment Analysis or Sentiment Classification and sometimes as Subjectivity Analysis (Cambria, Schuller, Xia, & Havasi, 2013). Most of the related work in opinion mining can be categorised into major categories like subjectivity classification, sentiment classification, review usefulness measurement, lexicon creation, opinion word, semantic annotations, product aspect extraction and many others (Ravi & Ravi, 2015). The objective of this paper is not to provide a survey of existing work on opinion mining, however, we will just give an overview of related work to give an idea about the latest progress being made in this field. Generally, approaches proposed for opinion mining can be classified into machine learning approaches and lexicon-based approaches (Missen, Boughanem, & Cabanac, 2013). However, there are some who have combined both types of approaches, hence called hybrid approach. The Machine Learning Approach (ML) (Lane, Clarke, & Hender, 2012; Ortigosa-Hernández et al., 2012; Walker et al., 2012) uses Machine Learning algorithms along with some linguistic features while the Lexicon-based Approach (Hu & Liu, 2004; Kim & Hovy, 2004) relies on a sentiment lexicon for this task (Medhat, Hassan, & Korashy, 2014). It has been observed that machine learning approaches have been generally more successful for opinion mining tasks and recently use of deep learning techniques (Glorot, Bordes, & Bengio, 2011; Paredes-Valverde, Colomo-Palacios, Salas-Zárate, & Valencia-García, 2017; Poria, Cambria, & Gelbukh, 2016; Tang, Wei, Qin, Liu, & Zhou, 2014; Zadeh, Chen, Poria, Cambria, & Morency, 2017).

As discussed above, there are many sub-tasks related to opinion mining. The semantic annotation of opinions is one of them. Semantic annotations are very important both for training machine learning approaches and for evaluating opinion mining methods. Unfortunately, there has been hardly any serious proposal attempts of appropriate annotation schemas until recently when SentiML (Di Bari, Sharoff, & Thomas, 2013), OpinionMiningML (Robaldo & Caro, 2013) and EmotionML (Schroder et al., 2011) were proposed. In this paper, we discuss, compare and identify the positives and negatives of these annotation schemes. Following this overview, we propose SentiML++, an extension of SentiML that addresses several shortcomings of the state of the art. As an extension of Missen, Attik, Coustaty, Doucet, & Faucher (2015) we also empirically evaluate SentiML and SentiML++ confirming the usefulness of SentiML++ over SentiML.

In Section 2, we discuss the existing annotation schemes, covering their positive and negative aspects. Section 3 presents our contribution SentiML++ in full details, detailing each addition using examples. Finally, Section 4 gives a demonstration of our annotation scheme before we conclude the paper and discuss future work in Section 5.

2. Annotation schemes

In this section, we discuss *SentiML*, *EmotionML* and *OpinionMiningML* and compare them as schemes for annotating opinion within text documents.

2.1. SentiML

The SentiML annotation schema (Di Bari et al., 2013) follows a conventional sentiment annotation style where one can find a target tag for the object about which sentiment is

being expressed while a modifier tag surrounds the expression containing sentiment information. However, it also adds to its vocabulary a much needed appraisal type tag which not only groups tags together but also describes the polarity of the compound expression. Besides this, SentiML is based on Appraisal Framework (AF) which is a strong linguistically-grounded theory. AF helps to define appraisal types (affect, judgments and appreciation) within the modifier tag which is another positive point to be noted in SentiML. With a very simple annotation scheme, SentiML does not require any specific skills. However, concerns can be raised about SentiML.

SentiML example

Throughout the article, we will use the following sentence as a running example:

> The U.S. State Department on Tuesday (KST) rated the human rights situation in North Korea "poor" in its annual human rights report, casting dark clouds on the already tense relationship between Pyongyang and Washington.

Relevant annotations in *SentiML* are given below:

```
<APPRAISALGROUP id="A0" fromID="T0" fromText="situation" toID="M0"
   toText="poor" orientation="negative"/>
<APPRAISALGROUP id="A1" fromID="M1" fromText="dark" toID="T1" toText="clouds"
   orientation="negative"/>
<APPRAISALGROUP id="A2" fromID="M2" fromText="tense" toID="T2"
   toText="relationship" orientation="negative"/>
<MODIFIER id="M0" start="201" end="205" text="poor" attitude="appreciation"
   orientation="negative" force="normal" polarity="unmarked"/>
<MODIFIER id="M1" start="250" end="254" text="dark" attitude="appreciation"
   orientation="negative" force="normal" polarity="unmarked"/>
<MODIFIER id="M2" start="277" end="282" text="tense" attitude="appreciation"
   orientation="negative" force="normal" polarity="unmarked"/>
<TARGET id="T0" start="175" end="184" text="situation" type="thing"
   orientation="neutral"/>
<TARGET id="T1" start="255" end="261" text="clouds" type="thing"
   orientation="ambiguous"/>
<TARGET id="T2" start="283" end="295" text="relationship" type="thing"
   orientation="neutral"/>
```

In the annotated sentence given above, three different phrases have been identified by using SentiML notation. Words like "situation", "clouds" and "relationship" have been identified as targets of the opinions expressed in this sentence. These targets have been identified on the phrase level and in the whole scenario, but the main target of the opinion i.e. North Korea has not been identified anywhere.

2.2. OpinionMiningML

OpinionMiningML (Robaldo & Caro, 2013) is an XML-based formalism that allows tagging of attitude expressions for features or objects as found in a textual segment. If analysed globally in terms of the related work on opinion mining, it can be categorised in the group of approaches targeting extraction of feature-based opinion expressions but its scope is limited to proposing an annotation schema. Its idea is to extract facets (i.e. features) from the text segment and form an ontology from these facets. Relations between these facets are defined using tags like <FEATURE-OF> and <SERVED-AT> while

expressions about these facets are defined separately using meta-tags like <APPRAISAL>, <OBSERVATIONS>, etc. This separation of marking facets, relations between facets and expressions with their polarity give a modular touch to this annotation scheme which can be considered a positive aspect of this formalism. However, the addition of a lot of meta-information about an expressive statement (ES) can be burdensome for annotators. Besides this, the structure introduced within the OpinionMiningML formalism is not straightforward and can be threatened by challenges for feature and relation extraction while developing an automatic tagger for this annotation scheme.

OpinionMiningML example

The example sentence is annotated below using the *OpinionMiningML* syntax:

```
<COMMENT id="1" ontologyreference="1">
<FRAGMENT id="1"> The U.S. State Department on Tuesday (KST) rated the human rights
   situation in North Korea "poor" </FRAGMENT>
<FRAGMENT id="2"> in its annual human rights report, casting dark clouds on the
   already tense relationship between Pyongyang and Washington.</FRAGMENT>
  <APPRAISAL polarity="negative" intensity="medium">
  <FACETREFERENCE>1</FACETREFERENCE>
  <FRAGMENTREFERENCE>1</FRAGMENTREFERENCE>
  <FRAGMENTREFERENCE>2</FRAGMENTREFERENCE>
  </APPRAISAL>
  <APPRAISAL polarity="negative" intensity="medium">
  <FACETREFERENCE>2</FACETREFERENCE>
  <FRAGMENTREFERENCE>1</FRAGMENTREFERENCE>
  <FRAGMENTREFERENCE>2</FRAGMENTREFERENCE>
  </APPRAISAL>
</COMMENT>
```

Note: *The annotation using OpinionMiningML requires a domain ontology for meta-tagging. This is why meta-tags such as <ONTOFACET> and <FACET> have not been defined in this example.*

2.3. EmotionML

EmotionML (Schroder et al., 2011) aims to make concepts from major emotion theories available in a broad range of technological contexts. Being informed by the affective sciences, EmotionML recognises the fact that there is no single agreed representation of affective states, nor of vocabularies to use. Therefore, an emotional state $< emotion >$ can be characterised using four types of descriptions: $< category >$, $< dimension >$, $< appraisal >$ and $< action - tendency >$. Furthermore, the vocabulary used can be identified.

EmotionML is aimed at three use cases: (1) Human annotation of emotion-related data, (2) automatic emotion recognition, and (3) generation of emotional system behaviour. In order to be suitable for all three domains, EmotionML is conceived as a "plug-in" language that can be used in different contexts. According to the W3C draft on EmotionML, one must refer to "one or more vocabularies to be used for representing emotion-related states". It is unfortunate that EmotionML specifications could not include a default vocabulary set because of disagreements in the community. Therefore, it was left at the discretion of users. "It is impossible to impose one single emotion vocabulary", Marc Schroeder,

editor of the EmotionML standard said (Schroder et al., 2011). "EmotionML is striking a balance, providing a carefully selected set of 'recommended' vocabularies" and documenting how to use them.

Things get more complicated, too. Moving beyond emotional states, there are "emotion dimension sets" that qualify emotions with attributes such as intensity and whether they are being experienced as positive or negative.

EmotionML example

We present hereby our running example annotated using the EmotionML syntax:

```
<emotionml xmlns="http://www.w3.org/2009/10/emotionml" xmlns:meta="http://
   www.example.com/metadata" category-set="http://www.w3.org/TR/emotion-voc/
   xml#occ-categories">
<info>
  <meta:doc>Example taken from annotation of SentiML
  </meta:doc>
</info>
The U.S. State Department on Tuesday (KST)
<emotion>
< category name="reproach"/>
  rated the human rights situation in North Korea "poor"
</emotion>
  in its annual human rights report
<emotion>
  < category name="disappointment"/>
  casting dark clouds
</emotion>
  on the already
<emotion>
  < category name="disappointment"/>
  tense relationship between Pyongyang and Washington.
</emotion>
</emotionML>
```

2.4. Comparison

In this section we give a comparison of annotations schemes from different perspectives.

- *Scope:* EmotionML, the W3C standard, is an effort to cover all aspects of emotions globally in all concerned fields while OpinionMiningML and SentiML limit themselves to domains of IR and NLP. In our view, SentiML has larger scope as compared to OpinionMiningML which limits itself only to feature-based sentiment analysis.
- *Complexity*: Because of its larger scope than the other two annotation schemes, EmotionML can be considered complex and less user-friendly. SentiML is much easier to use than OpinionMiningML because of the vocabulary of its annotation scheme which matches the vocabulary used in research work of this field, (i.e. concepts like holder, target, modifier, etc.).
- *Vocabulary*: EmotionML text annotation is equipped with new and broader vocabulary while targets of SentiML and OpinionMiningML are more specific. SentiML revolves around the concept of modifier and targets of the sentiments while OpinionMiningML's focus remains on sentiment relevant to features of the objects and equips itself with numerous meta-tags.

- *Structure*: The structure of all three annotation schemes follow XML based format but OpinionMiningML defines granularity better than the other annotation schemes by reaching up to the feature level.
- *Contextual Ambiguities:* While all three annotation schemes focus on defining semantics of the expression types, such as appreciation, suggestion etc., SentiML also tackles the issue of contextual ambiguities which is a big research challenge in the field of opinion mining.
- *Theoretical Grounds:* EmotionML is a W3C standard and has been recommended after years of discussion and debate between experts and stakeholders while SentiML is based on the Appraisal Framework, a strong linguistically-grounded theory. OpinionMiningML, in the contrary, has not been reported to have a theoretical basis.
- *Granularity Level:* Another distinction that can be observed while comparing these annotation schemes is their granularity level. EmotionML is found to be operating on the sentence level while OpinionMiningML and SentiML are rather focusing on sub-sentence levels. While it is mandatory to analyse on a sub-sentence (or word level) to find correct sentiments, the sentence is however a more logical unit of discourse.
- *Name:* One interesting observation that can be made of these three annotation schemes is their name. Quite interestingly the titles of all three use interchangeable words for opinions, i.e. opinion, emotion and sentiment.
- *Completeness*: Completeness (Oren, Moller, Scerri, Handschuh, & Sintek, 2006) is one of the most important features of annotation schemes. Completeness demands whether an annotation deals with all or most of the real world scenarios. Observing all of these annotations i.e. SentiML, EmotionML and OpinionMiningML in light of this particular characteristic, no annotation scheme seems to satisfy it. SentiML and EmotionML behave similarly by just annotating sentimental expressions while OpinionMiningML goes a step forward by targeting features of objects but fails to deal with contextual aspects of opinions, one of the major problems of opinion mining.
- *Flexibility*: EmotionML, knowing the challenges of different domains, gives freedom to use whatever vocabulary of emotional states suited to a particular domain. OpinionMiningML follows EmotionML in this regard and gives the choice of building a domain-based ontology as required, a possibility that SentiML lacks.

From this comparison, it can be concluded that SentiML has a larger scope and it is equipped with a more affordable vocabulary than in previous works (Cambria et al., 2013; Liu & Zhang, 2012; Missen et al., 2013; Pang & Lee, 2008). SentiML is based on Appraisal Framework (AF) which is a strong linguistically-grounded theory. Its simplicity is what will ensure a reliable process of annotation. Despite the advantages SentiML provides us, there are some limitations of SentiML that are discussed in next sub-section in detail.

2.5. Limitations of SentiML

Liu (2012) defines opinion more formally as:

> An opinion (or regular opinion) is a quintuple, $(e_i, a_{ij}, oo_{ijkl}, h_k, t_l)$, where e_i is the name of an entity, a_{ij} is an aspect of e_i, oo_{ijkl} is the orientation of the opinion about aspect a_{ij} of entity e_i,

h_k is the opinion holder, and t_l is the time when the opinion is expressed by h_k. The opinion orientation oo_{ijkl} can be positive, negative or neutral, or be expressed with different strength/intensity levels. When an opinion is on the entity itself as a whole, we use the special aspect GENERAL to denote it.

Taking this definition as a reference and mapping it to features of *SentiML*, it can be concluded that some aspects of opinion definition seem to be missing in SentiML. All such aspects and other important features lacking in SentiML are given below:

- Identifying holder of the opinion has been one of the very important tasks of the sentiment analysis field (Kim, Jung, & Myaeng, 2007; Seki, Kando, & Aono, 2009) and its importance marks it to be tagged in a more solid way especially in the schema of *SentiML* where other tags have been identified thoughtfully.
- While SentiML is equipped with a <TARGET> element unfortunately it operates only on sub-sentence level while fails to mark any sentence-level targets.
- As far as opinion orientation is concerned, SentiML deals with prior polarities while giving nothing to deal with contextual ambiguities. For example, it does not provide any support to recognise the topic of a sentence, hence fails to identify topic-based contextual ambiguities.
- SentiML fails to address annotation of informal language i.e. use of idioms or slang in text. Usually idioms totally change the semantics of text and are independent of words being used in them. In other words, idioms require sentence-level processing while SentiML works on phrase level.
- Flexibility and completeness are important characteristics of an annotation scheme (Oren et al., 2006) and unfortunately, *SentiML* fails to have both of these characteristics. Oren et al. define completeness of the annotations as how well an annotation capture the real-world situation. *SentiML* lacks many important elements of opinion mining in its toolbox.

Having discussed limitations of SentiML, it is time to introduce extension of SentiML i.e. SentiML++ equipped with concepts and structures tackling with limitations of SentiML.

3. SentiML ++

In this section, we define new elements as part of *SentiML*++, our proposed annotation scheme meant to overcome the weaknesses of SentiML. All the introduced elements are described in details with examples taken from the real-world corpus used by *SentiML* (Di Bari et al., 2013).

3.1. Target on sentence level

In the SentiML example quoted above, we have seen how it is unable to meet opinion annotation requirements. In the example, the actual target of the opinion cannot be identified at all. Therefore, we propose to mention concerned entities in the target on the sentence level as well as with the same syntax as it holds on the phrase level. Table 1 provides further details about the <TARGET> element.

Table 1. The <TARGET> element.

Annotation	<TARGET>
Definition	Defines the object about which sentiment has been expressed
Granularity Level	Sentence and Sub-sentence level
Attributes	• Required: ◦ target identity (id) ◦ start and end—The starting and ending position of the target in text ◦ text—The actual text of target i.e. how it appears in the given text ◦ type—The type of entity that is the target of the sentiment. It could have values like Thing, Country, Place, Organisation, Concept ◦ orientation—Semantic orientation of the target entity can be mentioned with values positive, negative or neutral • Optional: ◦ All attributes are Required
Occurrence	On the sentence level it will occur under <SENTENCE> while on the sub-sentence level it comes under <PHRASES>

Example:

```
<TARGET id="T0" start="108" end="117" text="North Korea" type="country"
orientation="neutral"/>
```

3.2. Identifying opinion holders

Identifying opinion holders has been a very important sub-task of opinion mining. In start, holder identification was more limited to news data collections because in reviews (e.g. of products or movies), the author of the opinion used to be considered as the holder of the opinion. Nowadays, these review sites have taken the form of online social networks and to avoid biased opinions and reviews, it has been necessary to identify the demographics of the opinion holder. This has also become more important also for news data so as to identify the actual entities influencing readers' opinions on the web. Therefore, we propose to add holder identification in SentiML++. Table 2 provides further details about the <HOLDER> element.

Example:

```
<HOLDER id="H0" start="33" end="53" text="U.S. State Department"
type="organization" orientation="neutral"/>
```

Table 2. The <HOLDER> element.

Annotation	<HOLDER>
Definition	Defines the object that expresses the sentiment
Level	Sentence and Sub-sentence level
Attributes	• Required: ◦ holder identity (id) ◦ start and end—The starting and ending position of the holder in text ◦ text—The actual text of the holder i.e. how it appears in the given text ◦ type—The type of entity that is the holder of the sentiment. It could have values like Thing, Country, Place, Organisation, Concept ◦ orientation—Semantic orientation of the holder entity can be mentioned with values positive, negative or neutral • Optional: ◦ All attributes are Required
Occurrence	On the sentence level it will occur under <SENTENCE> while on the sub-sentence level it comes under <PHRASES>

Table 3. The <TOPIC> element.

Annotation	<TOPIC>
Definition	Defines the topic of the sentence using a taxonomy
Level	Sentence
Attributes	• Required: ◦ topic identity (id) ◦ ref—It represents the referenced taxonomy used for defining the topic. It could be fixed for a given data collection ◦ chain—It lists the hierarchy of the topic of the sentence as described in a taxonomy • Optional: ◦ All attributes are Required
Occurrence	On the sentence level it will occur under <SENTENCE>

3.3. Marking topical domain with an ontology

As we have discussed above, the topical context of text highly affects the semantics of the opinion words. The same words can have different meanings in different contexts. Identification of opinion words and their polarity with respect to a given topic could help resolving contextual ambiguities. Therefore, we propose to take topic identification into account in SentiML++.

The topic is introduced to define the subject of each sentence. It can be based on any given taxonomy, such as, e.g. DMoz etc (Stoyanov & Cardie, 2008). The attribute Chain is supposed to list the chunk of taxonomy relevant to the topic of the given sentence. This very simple addition to SentiML could help solving many contextual problems of opinion mining. This, included with the orientation tag will aid in resolving contextual ambiguities of words as far as ambiguity arises because of the topic. Besides this, such annotations will also help creating domain level vocabularies of topical and semantic words. Table 3 provides further details about the <TOPIC> element.

Example:

```
<Topic id="TP1" ref="www.dmoz.org" chain="Society:Issues:Human Rights and
Liberties">
```

For example, the word “predictable” has a positive connotation if found in context of a product review talking about the camera’s function while the same word could be taken as negative when found in a movie review discussing the movie’s plot. A reasonable amount of data collection containing annotated documents from different domains could be very useful to reveal these ambiguous patterns. They would in turn be helpful to develop better opinion mining systems.

3.4. Dealing with emoticons, informal phrases

A good share of the opinions generally found on the web are expressed informally. This includes the use of emoticons and sarcastic phrases or cultural expressions (e.g. “bored to death”, “dressed to kill”, etc.) that could invert the semantics of the text surrounding them. Identification of such contexts could aid the automated detection of such opinion inversions.

With the increase in usage of online social networks, the usage of informal language is also increasing on the web. The use of idioms, emoticons (like ☺ ☹) and sarcastic phrases is not something that could be easily processed using conventional natural language processing techniques to reveal their semantics. Therefore, there is a need to go beyond the use of conventional techniques. One idea is to have data collections well-annotated with such text and then use machine learning to explore their context and predicting the actual semantics of such phrases. Therefore, we propose to annotate such textual segments using the structure explained by the example sentence *"You have a huge belly because you're a couch potato."*.

```
<informal id="I1" start="41" end="52" text="couch potato" support= "favors"
   orientation="negative BaseSentID= "S01" >
```

Such phrases could occur as part of a sentence or as an individual sentence on their own. This creates a reason for having an attribute "type" in markup "sentence" with two possible values i.e. "general" or "informal". If the informal text is part of a sentence then the value "general" is used while if it is a sentence in its own then the value "informal" is adequate. In case a sentence is categorized to the the "informal" type, it seems not necessary to annotate it with all the existing markup, e.g. target, modifier or appraisalgroup. However, it can be useful for another sentence, most probably the previous one, hence it can be added using the BaseSentID attribute. Table 4 describes the <INFORMAL> element in detail.

Note: In tables describing SentiML++ elements, the *sub-sentence* granularity level refers to the <PHRASES> in SentiML++ annotation.

3.5. SentiML++ example

In the previous section, we described the syntax of different markups proposed in SentiML ++. SentiML++ operates on both levels i.e. phrase as well as sentence level. In this section, we annotate the same sentence as an example using SentiML++. The annotation includes

Table 4. The <INFORMAL> element.

Annotation	<INFORMAL>
Definition	Defines the contextual polarity of an expression (idiomor emoticons) given in the text
Level	Sentence and Sub-sentence level
Attributes	• Required: ◦ identity (id) ◦ start and end—The starting and ending position of the informal text ◦ text—The actual text of the expression i.e. how it appears in the given text ◦ support—Describes how this particular expression semantically links to the given sentence or to another, i.e. whether it supports the semantics of the sentence or nullifies it or stays neutral ◦ orientation—The contextual polarity of the expression can be mentioned with values positive, negative or neutral ◦ BaseSentID—If the expression is part of a sentence then it describes the current sentence but if expression occurs independently of any sentence then it must be linking to the semantics of another sentence. In such case, it is necessary to provide the identity of the linked sentence • Optional: ◦ All attributes are Required
Occurrence	On the sentence level it will occur under <SENTENCE> while on the sub-sentence level it comes under <PHRASES>

the introduction of sentence-level markups like <HOLDER>, <TARGET>, <ORIENTATION>, <TOPIC> and <INFORMAL> while only <HOLDER> was introduced on the sub-sentence level. All of these markups come under the main markup of <SENTENCE> while sub-sentence level annotation comes under <PHRASES>. The markup <SENTENCE> includes one attribute called "type" with possible values "general" or "informal". The value "general" is used for ordinary sentences while the value "informal" is only used when the whole identified sentence is an idiom, a metaphor or an emoticon. When the value "informal" is used, only the <INFORMAL> markup plays its role while other markups are discarded because they are rendered meaningless. The <HOLDER> markup was introduced on the sub-sentence level to identify the holders even at this smaller granularity, if needed.

```
<SENTENCE ID=0 type="general">
  <HOLDER id="H0" start="33" end="53" text="U.S. State Department"
   type="organization" orientation="neutral"/>
  <TARGET id="T0" start="108" end="117" text="North Korea" type="country"
   orientation="neutral"/>
  <ORIENTATION polarity="negative">
  <TOPIC id="1" ref="www.dmoz.org" chain="Society:Issues:Human Rights and
   Liberties">
  <PHRASES>
  <APPRAISALGROUP id="A0" fromID="T0" fromText="situation" toID="M0"
   toText="poor" Holder="H0" orientation="negative"/>
  <APPRAISALGROUP id="A1" fromID="M1" fromText="dark" toID="T1" toText="clouds"
   Holder="H1" orientation="negative"/>
  <APPRAISALGROUP id="A2" fromID="M2" fromText="tense" toID="T2"
   toText="relationship" Holder="H2" orientation="negative"/>
  <MODIFIER id="M0" start="201" end="205" text="poor" attitude="appreciation"
   orientation="negative" force="normal" polarity="unmarked"/>
  <MODIFIER id="M1" start="250" end="254" text="dark" attitude="appreciation"
   orientation="negative" force="normal" polarity="unmarked"/>
  <MODIFIER id="M2" start="277" end="282" text="tense" attitude="appreciation"
   orientation="negative" force="normal" polarity="unmarked"/>
  <TARGET id="T1" start="175" end="184" text="situation" type="thing"
   orientation="neutral"/>
  <TARGET id="T2" start="255" end="261" text="clouds" type="thing"
   orientation="ambiguous"/>
  <TARGET id="T3" start="283" end="295" text="relationship" type="thing"
   orientation="neutral"/>
  <HOLDER id="H1" start="33" end="53" text="U.S. State Department"
   type="organization" orientation="neutral"/>
  <HOLDER id="H2" start="33" end="53" text="U.S. State Department"
   type="organization" orientation="neutral"/>
  <HOLDER id="H3" start="33" end="53" text="U.S. State Department"
   type="organization" orientation="neutral"/>
  </PHRASES>
  </SENTENCE>
```

It must be noted that <APPRAISALGROUP> in SentiML++ links modifier, target and holder identified at the sub-sentence level. This is in contradiction with SentiML (Di Bari et al., 2013), where only modifier and target are linked. <Holder> and <Target> elements are found on both levels i.e. sentence and sub-sentence level. Natural language processing techniques such as syntactic parsing can be helpful in identifying these elements on both granularity levels.

4. SentiML Vs SentiML++: evaluation

4.1. Data collection

For evaluating positive role of SentiML++, we use the same data collection as provided by contributors of SentiML. This data collection (Di Bari et al., 2013) contains text documents from following three sources with a total number of almost 7,000 words:

- Political Speeches: Mainly American presidents' addresses as collected from the web.
- Talks: TED (Technology, Entertainment, Design) talks on a wide range of topics under the slogan "ideas worth spreading".
- Political Speeches: Belonging to the "human rights" domain of the MPQA opinion corpus (Wilson, 2008). This corpus contains texts with manual annotations of opinions and private states such as beliefs, emotions, sentiments and speculations.

First challenge in this regards is to annotate this data collection with elements of SentiML++. Next subsection describes how this annotation work is performed.

4.2. SentiML++ annotation

There are six basic steps to annotation with SentiML++ that involves:

- identifying sentences in a document,
- identifying holders of opinion in a sentence,
- identifying target of opinion in a sentence,
- identifying sentiment of the opinion expressed in a sentence,
- identifying the topic of the given sentence,
- identifying holder of opinion on phrase level.

We recruited five participants to annotate the data collection with provided guidelines. All participants are computer science graduates with more than 5 years of experience of teaching in a university.

4.3. Annotation guidelines

All annotators are given following guidelines while annotating a given document:

- *Sentence Polarity* Mark a sentence as positive (or negative) if it shows a positive (or negative) attitude towards its target.If you fail to conclude its semantic orientation, mark it as a neutral sentence.
- *Opinion Holder or Target* Holders of an opinion (if exists) can be traced in subject(s) of a sentence. Similarly, you will often find target of an opinion as object of a sentence. However, if you fail to identify one or both of them, mark nothing in holder and target elements. This is true for both on sentence as well as on phrase level.
- *Sentence Topic* Topic of a sentence can be traced using a web directory like DMoz. Mention the nearest chain that you think relates to content of the sentence.

Fleiss Kappa (Fleiss, 1971) agreement score of 0.82 (overall) shows that data collection was annotated without too many disagreements. Mostly, annotators agreed on identifying sentiment (Fleiss Kappa score 0.83), holder and target of opinions (Fleiss Kappa score 0.85) but they differ on topic identification of the sentence using DMoz (Fleiss Kappa score 0.78).

4.4. Evaluation

Considering the size of annotated data collection for both SentiML and SentiML++, we generate 15 queries (see Table 5) for which both can be evaluated. Queries are generated with focus on natural language style of questioning. We compare results of SentiML and SentiML++ by segmenting queries and by comparing the returned results by both types of annotations.

We formally experiment with fifteen queries by selecting a standard metric of evaluation. Choosing a standard evaluation metric in this particular case is very tricky because we do not have lot of alternatives for relevance. Precision and Recall behave the same when the number of relevant documents in the collection and the total documents being retrieved are one and the same as in our case. In other words, our annotation should perform like an ideal system because we have only one relevant segment within the given corpus. Therefore, using R-Precision (Manning, Raghavan, & Schutze, 2008) a metric of evaluation is a suitable choice in this case giving us a value of 1 for an ideal annotation while a value of zero for the annotation failed to perform well. R-Precision can be defined as:

$$\mathrm{R-Precision} = \frac{r}{|R|},$$

is the total number of relevant documents in the collection. In an ideal system R can take a value of 1 (in our case most of the time). This is why we consider R-precision a suitable choice for evaluation. Hence, we re-perform our experiments with total 15 queries. R-Precision is computed for each query for both SentiML and SentiML++ annotations. This query-based evaluation (see Table 6) significantly confirms that SentiML++ is better than SentiML (with $p-$value $< .05$) at answering natural language queries with mean R-Precision score of 0.26 for SentiML against 1.0 for SentiML++. SentiML lacks important

Table 5. Set of Natural Language Queries Generated.

Query 1	Who blamed North Korean Authorities ?
Query 2	What is report about?
Query 3	How many times the report said something negative about North Korea?
Query 4	Who runs the greatest anti-poverty program?
Query 5	What IFRC says about human rights situation in Iran?
Query 6	Which entity named the US report a “groundless plot”?
Query 7	On what issue president Kim made a very positive statement?
Query 8	Who are detained arbitrarily?
Query 9	What causes detention of citizens?
Query 10	Who said that “Terrorists believe that anything goes in the name of their cause”.
Query 11	In IHRC declaration who was target of injustice?
Query 12	Which organization called on Israel to stop bombing?
Query 13	Analysts have warned about what?
Query 14	What is the purpose of Mary Robinson visit to China?
Query 15	What President Obama had to say about violence?

Table 6. Evaluation Results of SentiML and SentiML++ with R-Precision.

Query	SentiML	SentiML++
Query 1	0	1
Query 2	0	1
Query 3	0	1
Query 4	1	1
Query 5	0	1
Query 6	0	1
Query 7	0	1
Query 8	1	1
Query 9	0	1
Query 10	0	1
Query 11	1	1
Query 12	0	1
Query 13	1	1
Query 14	0	1
Query 15	0	1
Mean	0.26	1**

elements like holder and sentence-based orientations which makes it difficult to answer such queries. Therefore, SentiML will have to seek support of natural language processing to answer these queries while SentiML++ does it with the help of its elements.

5. Conclusions and future work

In this work, we proposed *SentiML++*, an extension of SentiML for automatic processing of opinion data. We proposed to add target (on sentence level), holder, topic, and informal sentence identification as part of *SentiML++*. Our *SentiML++* adds flexibility to SentiML by giving freedom of choice for a general taxonomy when annotating the topic of a sentence. As part of our future work, we plan to model a precise and compact annotation model for sentiment identification beyond the scope of SentiML or any of the other annotation schemes as a reference. The idea is to propose a suitable model for the semantic web such that the state-of-the-art semantic web tools can be leveraged to exploit its semantics. Further, in the future annotation models, we plan to include objects, sentiments and time as well as link them using semantic relations.

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References

Cambria, E., Schuller, B., Xia, Y., & Havasi, C. (2013). New avenues in opinion mining and sentiment analysis. *IEEE Intelligent Systems*, *28*(2), 15–21.

Di Bari, M., Sharoff, S., & Thomas, M. (2013). Sentiml: Functional annotation for multilingual sentiment analysis. *Proceedings of the 1st international workshop on collaborative annotations in shared environment: Metadata, vocabularies and techniques in the digital humanities* (pp. 15:1–15:7). New York, NY: ACM. doi:10.1145/2517978.2517994

Fleiss, J. L. (1971). Measuring nominal scale agreement among many raters. *Psychological Bulletin*, *76*(5), 378–382.

Glorot, X., Bordes, A., & Bengio, Y. (2011). Domain adaptation for large-scale sentiment classification: A deep learning approach. *Proceedings of the 28th international conference on machine learning (ICML-11)*, Bellevue, WA (pp. 513–520).

Hu, M., & Liu, B. (2004). Mining and summarizing customer reviews. *Proceedings of the tenth ACM SIGKDD international conference on knowledge discovery and data mining*, Seattle, WA, (pp. 168–177).

Kim, S.-M., & Hovy, E. (2004). Determining the sentiment of opinions. *Proceedings of the 20th international conference on computational linguistics*, Geneva (p. 1367).

Kim, Y., Jung, Y., & Myaeng, S. H. (2007). Identifying opinion holders in opinion text from online newspapers. In *Proceedings of the international conference on granular computing* (pp. 699–702). Berlin, Heidelberg: Springer-Verlag.

Lane, P. C. R., Clarke, D., & Hender, P. (2012). On developing robust models for favourability analysis: Model choice, feature sets and imbalanced data. *Decision Support Systems*, *53*(4), 712–718.

Liu, B. (2012). *Sentiment analysis and opinion mining*. Morgan & Claypool Publishers. doi:10.2200/S00416ED1V01Y201204HLT016

Liu, B., & Zhang, L. (2012). A survey of opinion mining and sentiment analysis. In C. C. Aggarwal & C. Zhai (Eds.), *Mining text data* (pp. 415–463). Springer US. doi:10.1007/978-1-4614-3223-4_13

Manning, C. D., Raghavan, P., & Schutze, H. (2008). *Introduction to information retrieval*. Cambridge: Cambridge University Press.

Medhat, W., Hassan, A., & Korashy, H. (2014). Sentiment analysis algorithms and applications: A survey. *Ain Shams Engineering Journal*, *5*(4), 1093–1113.

Missen, M. M. S., Attik, M., Coustaty, M., Doucet, A., & Faucher, C. (2015). Sentiml++: An extension of the sentiml sentiment annotation scheme. *International semantic web conference*, Portorož (pp. 91–96).

Missen, M. M. S., Boughanem, M., & Cabanac, G. (2013). Opinion mining: Reviewed from word to document level. *Social Network Analysis and Mining*, *3*(1), 107–125. doi:10.1007/s13278-012-0057-9

Oren, E., Moller, K., Scerri, S., Handschuh, S., & Sintek, M. (2006). *What are semantic annotations?* (Tech. Rep.). DERI Galway. Retrieved from http://www.siegfried-handschuh.net/pub/2006/whatissemannot2006.pdf

Ortigosa-Hernández, J., Rodríguez, J. D., Alzate, L., Lucania, M., Inza, I., & Lozano, J. A. (2012). Approaching sentiment analysis by using semi-supervised learning of multi-dimensional classifiers. *Neurocomputing*, *92*, 98–115.

Pang, B., & Lee, L. (2008, January). Opinion mining and sentiment analysis. *Foundations and Trends*^{TM} *in Information Retrieval*, *2*(1–2), 1–135. doi:10.1561/1500000011

Paredes-Valverde, M. A., Colomo-Palacios, R., Salas-Zárate, M. D. P., & Valencia-García, R. (2017). Sentiment analysis in Spanish for improvement of products and services: A deep learning approach. *Scientific Programming*, *2017*. doi:10.1155/2017/1329281

Poria, S., Cambria, E., & Gelbukh, A. (2016). Aspect extraction for opinion mining with a deep convolutional neural network. *Knowledge-Based Systems*, *108*, 42–49.

Ravi, K., & Ravi, V. (2015). A survey on opinion mining and sentiment analysis: Tasks, approaches and applications. *Knowledge-Based Systems*, *89*, 14–46.

Robaldo, L., & Caro, L. D. (2013). Opinionmining-ml. *Computer Standards and Interfaces*, *35*(5), 454–469. Retrieved from http://dblp.uni-trier.de/db/journals/csi/csi35.html#RobaldoC13

Schroder, M., Baggia, P., Burkhardt, F., Pelachaud, C., Peter, C., & Zovato, E. (2011). Emotionml—an upcoming standard for representing emotions and related states. *Proceedings of affective computing and intelligent interaction*, Memphis, TN. Springer.

Seki, Y., Kando, N., & Aono, M. (2009). Multilingual opinion holder identification using author and authority viewpoints. *Journal of Information Processing and Management*, *45*(2), 189–199.

Stoyanov, V., & Cardie, C. (2008, May). Annotating topics of opinions. In N. C. C. Chair, et al. (Eds.), *Proceedings of the sixth international conference on language resources and evaluation (LREC'08)*. Marrakech, Morocco: European Language Resources Association (ELRA). Retrieved from http://www.lrec-conf.org/proceedings/lrec2008/

Tang, D., Wei, F., Qin, B., Liu, T., & Zhou, M. (2014). Coooolll: A deep learning system for twitter sentiment classification. *Proceedings of the 8th International Workshop on Semantic Evaluation (SemEval 2014)*, Dublin, Ireland (pp. 208–212).

Walker, M. A., Anand, P., Abbott, R., Tree, J. E. F., Martell, C., & King, J. (2012). That is your evidence?: Classifying stance in online political debate. *Decision Support Systems*, *53*(4), 719–729.

Wilson, T. A. (2008). *Fine-grained subjectivity and sentiment analysis: Recognizing the intensity, polarity, and attitudes of private states* (Unpublished doctoral dissertation). Retrieved from http://d-scholarship.pitt.edu/7563/

Zadeh, A., Chen, M., Poria, S., Cambria, E., & Morency, L.-P. (2017). Tensor fusion network for multimodal sentiment analysis. *Conference on empirical methods in natural language processing*, Copenhagen (pp. 1103–1114).